Cloud of Ellipsoids: Local Geometric Modelling of DINOv2 Embeddings for Unsupervised Anomaly Detection

Anonymous submission

Abstract

This paper investigates how normal image representations can be modelled in an unsupervised embedding space using multiple local multivariate regions rather than a single global distribution. DINOv2 embeddings are extracted from the 15 object and texture categories of MVTec AD, and a Cloud of Ellipsoids method is proposed to represent normal variation using locally fitted covariance-aware regions. Each test sample is scored according to its relationship with the learned ellipsoids, providing both an anomaly score and a local geometric interpretation of how it relates to the normal embedding space. The method is compared with a global Mahalanobis-distance baseline. Across the dataset, the proposed approach achieves a mean AUROC of 0.934 compared with 0.958 for Mahalanobis, indicating that the additional flexibility of multiple local regions does not improve overall anomaly-detection performance. Bootstrap evaluation further shows greater sensitivity of the ellipsoidal method to changes in the available training samples. These results suggest that local geometric modelling may be more effective when used to refine a stable global representation rather than constructing the normal embedding space entirely from independently fitted local regions.

1. Introduction

In conventional classification, a sample $x \in \mathbb{R}^n$ is assigned to one of K classes based on properties shared by samples belonging to that class. However, representing a class becomes more difficult when its distribution is heterogeneous or multimodal. A single global decision region may then become overly broad, covering areas of the feature space that do not accurately represent the observed samples. This is particularly relevant to unsupervised anomaly detection, where defective examples may be unavailable during training and the normal distribution must be modelled without prior knowledge of possible anomalies.

Representation-learning models map visually complex images into embedding spaces in which similarity can be analysed geometrically. However, normal samples are not necessarily well represented by a single centroid, sphere, or global covariance estimate. Different normal appearances may occupy separate or elongated regions, causing a global model either to exclude valid normal samples or include unsupported areas of the embedding space.

The *Cloud of Spheres* method proposed by T. Dias and P. Amaral [1] addresses complex class structure using multiple connected spherical regions rather than a single global region. Its supervised formulation constructs a connected set of spheres around the target class through mixed-integer nonlinear optimisation while minimising the number of required regions. This provides an explicit geometric representation capable of describing nonlinear and non-convex class boundaries.

This paper extends this approach by proposing a *Cloud of Ellipsoids* for unsupervised anomaly detection. Rather than constraining each local region to be isotropic, ellipsoids adapt to the covariance and local geometry of the representation space, allowing different variation and orientation along each axis. Normal space can therefore be represented by multiple locally fitted regions without requiring defective training examples.

DINOv2 [2] is used to generate the representation space, despite not being trained specifically for anomaly detection, with the proposed method evaluated on the 15 object and texture categories of MVTec AD [3]. The primary objective is to determine whether the natural variation within normal images forms local structure that can be captured effectively using multiple ellipsoidal regions, particularly where a single global distribution provides an inadequate description of the embedding space. This local representation also provides a more explicit geometric interpretation of normality by relating each sample to specific regions of the learned space rather than only to a single global score. Performance is compared with simpler global distance-based anomaly-detection approaches at the image level.

2. Related Work

Two established anomaly-detection approaches evaluated on MVTec AD are PaDiM [4] and PatchCore [5]. Both use pretrained convolutional neural networks (CNNs) to extract representations from normal training images, but differ in how the resulting normal feature space is modelled.

Mahalanobis distance provides a simple distribution-based approach to anomaly detection by measuring the distance of a sample from a reference distribution while accounting for feature covariance [6]. It is particularly relevant to this work because it provides the baseline scoring method for the DINOv2 embeddings and is closely related to the covariance-aware geometry used by the proposed ellipsoidal regions. Whereas a global Mahalanobis model represents normality using a single covariance structure, the proposed approach instead models multiple local regions.

2.1. Representation-based Anomaly Detection

PaDiM models normal features independently at each spatial position using multivariate Gaussian distributions. Features from multiple levels of a pretrained CNN are combined, and at inference the Mahalanobis distance between each test feature and its corresponding normal distribution is used to identify anomalous regions. To reduce computational cost, PaDiM also applies random feature selection, with the authors showing that strong anomaly-detection performance can be retained using a reduced representation. However, retaining similar AUROC does not establish which properties of the original representation have been preserved or discarded, which is relevant when interpretability of the embedding geometry is also of interest.

PatchCore instead represents normality using a memory bank of local features extracted from intermediate layers of a pretrained CNN. Test patches are compared with nearby normal representations in this memory, with greater distance indicating a more anomalous feature. Unlike PaDiM, PatchCore therefore retains representative normal examples rather than explicitly fitting a probability distribution.

PaDiM and PatchCore were not reimplemented here as this work focuses on embedding-space geometry rather than patch-level localisation; comparison is limited to Mahalanobis as the most directly related global baseline.

2.2. Geometric Modelling with Cloud of Spheres

The original *Cloud of Spheres* algorithm [1] represents complex binary class boundaries using multiple spherical regions rather than a single global decision region. It is formulated as a supervised optimisation problem in which positive and negative training samples constrain the construction of the cloud, while the number of required spheres is minimised. The resulting geometric representation allows nonlinear and non-convex class structures to be described using a collection of simpler local regions.

Several assumptions of the original formulation do not directly transfer to unsupervised anomaly detection. *Cloud of Spheres* uses examples from both classes to constrain its decision boundary, whereas only normal training samples are available in the setting considered here. The location and structure of anomalous regions are therefore unknown during construction of the normal representation.

The original method also assumes that samples belonging to a class form a connected manifold and consequently encourages neighbouring spheres to form a connected cloud. In an unsupervised anomaly-detection setting, however, only normal samples are available, providing no information about the location or structure of anomalous regions between observed areas of normality. Enforcing connectivity may therefore incorrectly treat unsupported regions of the embedding space as normal simply because they lie between separate normal clusters. This is particularly restrictive for learned visual representations, where normal samples may occupy multiple distinct regions of a high-dimensional embedding space. In addition, spherical regions assume approximately isotropic local structure and cannot adapt their orientation to directional variation within the representation.

These limitations motivate the extension investigated in this work: an unsupervised multi-region model that does not require connectivity between normal regions and replaces spherical boundaries with covariance-adaptive ellipsoids.

3. Method

DINOv2 ViT-S/14 is used to extract the 384-dimensional [CLS] embeddings from the normal training images, which form the representation space modelled by the algorithm. Images are resized to 224×224 pixels and normalised using the ImageNet[7] channel statistics employed by the DINOv2[2] preprocessing pipeline, with mean [0.485, 0.456, 0.406] and standard deviation [0.229, 0.224, 0.225] for the red, green, and blue channels respectively. Ellipsoids are fitted using the standard MVTec AD[3] training split, which contains only normal samples, and evaluated on the corresponding test split containing both normal and anomalous samples. Ellipsoids are constructed sequentially from the remaining unassigned embeddings. After a candidate region is fitted and accepted, its assigned embeddings are consumed, and the process repeats until the training set has been represented.

3.1. Baseline

A global Mahalanobis model is fitted independently for each MVTec AD category using its normal training embeddings. The centroid μ is calculated from all available training embeddings, after which SVD[8] is applied to the centred embedding matrix. The resulting singular values are converted to covariance eigenvalues, with numerically negligible directions removed using a machine-precision tolerance. This produces a single covariance-aware representation of the normal embedding space for each category.

At inference, a test embedding is projected onto the retained principal directions and its Mahalanobis distance from the global centroid is calculated. A regularisation term of 1×10^{-6} is added to the retained eigenvalues for numerical stability. Any component of the embedding lying outside the retained subspace is additionally penalised using the same regularisation term. The resulting distance is used directly as the anomaly score, with larger values indicating greater deviation from the fitted normal distribution. The score is defined as

$$d(x) = \sqrt{\sum_i \frac{\left(v_i^T(x-\mu)\right)^2}{\lambda_i + \varepsilon} + \frac{\|r(x)\|_2^2}{\varepsilon}}$$

where v_i and λ_i denote the retained principal directions and corresponding covariance eigenvalues, respectively, and $r(x)$ is the residual component of $x - \mu$ lying outside the retained subspace.

3.2. The Starting Position

The algorithm begins by identifying the densest local region of the remaining normal embedding space using K-nearest neighbours (KNN)[9]. The embedding with the smallest mean distance to its K neighbours is selected as the initial centre.
Because MVTec AD categories contain different numbers of training samples, K is defined as a proportion of the available embeddings rather than a fixed count. Candidate values of 2.5%, 5%, and 10% were evaluated, balancing sensitivity to local noise at small K against overly broad initial regions at large K. A value of 5% was selected as a practical compromise across categories, although the most suitable neighbourhood size varied between categories.

3.3. Ellipsoid Fitting

Given a set of normal embeddings assigned to a candidate region, an ellipsoid is fitted to describe its local geometry. The centre μ is calculated as the weighted mean of the assigned embeddings. The samples are then centred around μ and scaled according to their weights.

Let X_w denote the resulting weighted centred data matrix. Singular Value Decomposition (SVD)[8] is applied as $X_w = U \Sigma V^T$.
The columns of V define the principal directions of the region, while the singular values in Σ are used to derive the variance λ_i associated with each supported axis. Directions with negligible variance are removed, allowing the ellipsoid to retain only the dimensional structure supported by the local samples.

The boundary of the ellipsoid is determined from the distances of its assigned embeddings. The boundary threshold τ is fitted as the maximum weighted squared ellipsoidal distance among the samples supporting the region. Fully weighted samples therefore determine the fitted extent of the ellipsoid, while downweighted samples exert progressively less influence on its boundary. Membership of a point x is then determined using its squared ellipsoidal distance.

$$\sum_i \frac{\left(v_i^T (x - \mu)\right)^2}{\lambda_i + \varepsilon} \leq \tau$$

Here, v_i and λ_i denote the principal direction and variance associated with axis i, while ε is a small regularisation term used for numerical stability set as 1×10^{-4}.
This covariance-aware representation allows each region to adapt to the locally observed shape of the embedding distribution rather than assuming equal variation in every direction.

3.4. Low-Support Ellipsoid Fitting

As normal embeddings are progressively assigned to ellipsoids, later candidate regions are constructed from fewer remaining samples. With very small sample sizes, the estimated covariance becomes increasingly unstable, which can produce poorly defined principal directions and unreliable growth. Covariance estimators such as Ledoit-Wolf[10] can stabilise small-sample estimates through shrinkage towards a more isotropic covariance structure. However, this is undesirable here because it can introduce variance in directions that are weakly supported by the local samples, potentially encouraging subsequent ellipsoid growth into unsupported regions of the embedding space.

Based on validation analysis of ellipsoid size and covariance structure, including the ratio between the largest and smallest eigenvalues and the proportion of variance explained by the dominant principal component, candidate regions containing fewer than five samples were treated as low-support regions. Rather than relying entirely on their own covariance estimate, these regions borrow covariance structure from a previously fitted ellipsoid while progressively retaining more of their own geometry as additional local samples become available.

Support selection is performed in two stages. First, the candidate set is restricted to the five previously fitted ellipsoids whose centres are nearest to the candidate centre under Euclidean distance. This preserves locality in the original DINOv2 embedding space before geometric similarity is considered.

For a singleton candidate, no local covariance orientation can be estimated. The support ellipsoid is therefore selected using both embedding similarity and the alignment between the candidate direction and the existing support geometry. Let c denote the candidate embedding and μ_s the centre of a support ellipsoid. The unit direction from the support towards the candidate is

$$d = \frac{c - \mu_s}{\|c - \mu_s\|}.$$

Embedding similarity is measured using cosine similarity,

$$S_{\text{embed}} = \frac{c^T \mu_s}{\|c\| \|\mu_s\|}.$$

Geometric similarity measures how closely d aligns with the principal axes $v_{s,i}$ of the support ellipsoid. Each alignment is weighted by the corresponding axis length,

$$S_{\text{shape}} = \frac{\sum_i \left|d^T v_{s,i}\right| \sqrt{\lambda_{s,i} + \varepsilon}}{\sum_i \sqrt{\lambda_{s,i} + \varepsilon}}.$$

The final singleton support score is the mean of the two components,

$$S_{\text{support}} = \frac{S_{\text{embed}} + S_{\text{shape}}}{2}.$$

For low-support candidates containing more than one sample, some local covariance structure can already be estimated. Support similarity is therefore defined as the mean of the singular values of $V_c^T V_s$, which measures alignment between the candidate and support principal subspaces.

After selecting the highest-scoring local support ellipsoid, its covariance structure is blended with that of the candidate according to $C_b = \alpha C_c + (1 - \alpha) C_s$, where $\alpha = \min\left(1, \frac{n}{5}\right)$ Consequently, the contribution of the borrowed covariance decreases as additional local evidence becomes available, with the candidate covariance used independently once $n \geq 5$.

3.5. Ellipsoid Growth

The initial KNN neighbourhood provides a dense starting region, but its size should not directly determine the final ellipsoid. A growth stage is therefore used to expand each candidate region and incorporate nearby normal embeddings. When new points are enclosed, the centroid and covariance of the region are recalculated before growth continues. This process repeats until no additional embeddings can be incorporated.

Growth is *non-uniform* across the ellipsoid axes. Rather than applying the same scale to every direction, growth is proportional to the variance already observed along each principal axis. Let λ_i denote the variance associated with axis i and $\lambda_{\max}$ the largest eigenvalue. The relative axis variance is

$$r_i = \max\left(\frac{\lambda_i}{\lambda_{\max}}, r_{\min}\right).$$

Given a global growth factor g, the scale applied to axis i is $a_i = 1 + (g - 1) r_i, \quad \lambda_{i'} = \lambda_i a_i^2.$ Here, $r_{\min}$ prevents axes with very small but non-zero variance from receiving no growth. Consequently, an axis with $r_i = 1$ receives the full growth factor g, while lower-variance directions expand more conservatively.

3.6. Candidate Cleaning

As ellipsoids are constructed sequentially, embeddings assigned to an earlier region are removed from consideration when fitting subsequent regions. However, a newly fitted candidate may still geometrically expand into a previously assigned embedding. Candidate cleaning is therefore introduced to constrain this encroachment while preserving as much of the candidate's local covariance structure as possible.

Rather than immediately removing an embedding from the candidate neighbourhood, each candidate embedding is assigned a weight $w_i \in [0, 1]$. These weights are normalised before the weighted centroid and covariance structure are estimated using the fitting procedure described above. A weight of one represents full contribution to the fitted geometry, while a weight of zero removes that embedding from the covariance estimate.

When a previously assigned embedding x_s lies inside the candidate ellipsoid, its contribution along each retained principal axis is evaluated as

$$c_{i(x_s)} = \frac{\left(v_i^T (x_s - \mu)\right)^2}{\lambda_i}.$$

The axis with the greatest contribution is $i^* = \arg\max_i c_{i(x_s)}.$

Reducing the candidate variance along this direction provides the most direct means of moving the conflicting point towards the ellipsoid boundary. The candidate embedding contributing most strongly to this axis is therefore selected as

$$j^* = \arg\max_j \frac{\left(v_{i^*}^T (x_j - \mu)\right)^2}{\lambda_{i^*}}.$$

Rather than immediately discarding x_{j^*}, its weight is reduced using binary search. After each update, the ellipsoid is refitted and the conflicting embedding is tested again. The largest weight that removes the encroachment is retained, preserving as much of the original neighbourhood as possible. If no positive weight resolves the conflict, the embedding weight is reduced to zero and the candidate is refitted without its contribution.

3.7. Scoring

At inference, each test embedding is evaluated against every ellipsoid in the learned cloud. From ellipsoid j, the boundary margin is defined as the squared ellipsoidal distance from the sample minus the fitted threshold, $m_{j(x)} = d^2_{j(x)} - \tau_j$, the anomaly score is the minimum margin across all ellipsoids $s(x) = \min_j m_{j(x)}$ with $j^* = \arg\min_j m_{j(x)}$ identifying the corresponding normal region. Negative scores indicate that the sample lies within at least one ellipsoid, while positive scores indicate that it lies outside every modelled region of normality. Increasing positive margin therefore represents increasing deviation from the learned normal space.

Hyperparameter	Value
K (init. neighbourhood)	5%
Low-support threshold	5
Local support candidates	5
Growth factor g	1.1

Hyperparameter	Value
Ellipsoidal Reg. ε	1×10^{-4}
Mahalanobis Reg. ε	1×10^{-6}
$r_{\min}$	1×10^{-4}
Blending weight α	$\min\left(1, \frac{n}{5}\right)$

4. Results

Results are reported using image-level AUROC [11]. AUROC measures the probability that a randomly selected anomalous sample receives a higher anomaly score than a randomly selected normal sample, providing a threshold-independent measure of how well the two groups are ranked.

Results Table (AUROC)

Category	Ellip.	Mahal.
bottle	0.997	1.0
cable	0.883	0.945
capsule	0.874	0.941
carpet	0.972	0.98
grid	0.97	0.983
hazelnut	0.931	0.949
leather	1.0	1.0
metal_nut	0.935	0.983
pill	0.91	0.95
screw	0.808	0.802
tile	0.999	1.0
toothbrush	0.958	0.972
transistor	0.891	0.935
wood	0.902	0.944
zipper	0.983	0.987

Overall, the proposed ellipsoidal algorithm performs worse than the Mahalanobis baseline, achieving a mean AUROC of 0.934 compared with 0.958 for Mahalanobis. This corresponds to a reduction of 0.024 in mean AUROC, indicating that the additional flexibility introduced by modelling multiple ellipsoidal regions does not generally improve anomaly separation across the dataset. Performance varies between categories, however, and screw is a notable exception, where the ellipsoidal approach improves upon the baseline. This initially suggests a possible category-specific benefit, although the bootstrap analysis below examines whether this improvement is robust.

However, these benchmark results are based on a single train/test configuration and therefore do not indicate how stable either method is to variation in the available samples. To assess robustness, training and test variation were evaluated separately using 1,000 bootstrap resamples. For the training bootstrap, normal training embeddings were resampled with replacement and both models were refitted before evaluation on the original test set. For the test bootstrap, normal and anomalous test embeddings were resampled with replacement while the fitted models were held fixed. Within each bootstrap iteration, both methods were evaluated using the same resampled data, allowing paired differences in AUROC to be calculated. Reported 95% intervals correspond to the 2.5th and 97.5th percentiles of the bootstrap distributions.

Across the training bootstraps, the difference between the two methods becomes more pronounced. The ellipsoidal algorithm decreases to a mean AUROC of 0.924 (95% CI [0.903, 0.944]), while the Mahalanobis baseline remains comparatively stable at 0.957 (95% CI [0.944, 0.967]), producing a mean difference of −0.033 (95% CI [−0.054, −0.011]).

In contrast, performance of the ellipsoidal approach remains more stable across the test bootstraps, maintaining a mean AUROC of approximately 0.934 (95% CI [0.887, 0.971]). The Mahalanobis baseline increases to approximately 0.964 (95% CI [0.927, 0.986]), producing a mean paired difference of −0.030 (95% CI [−0.059, −0.001]) and further increasing the performance gap between the two methods.

The improvement previously observed for screw does not remain stable under bootstrap resampling. While the ellipsoidal approach slightly outperformed the Mahalanobis baseline for screw in the original benchmark, this advantage is not maintained across either the training or test bootstraps, where the ellipsoidal method instead performs worse than the baseline on average.

In terms of computational cost, Mahalanobis is expected to be faster to fit, as it models a single global distribution, whereas the proposed method constructs multiple ellipsoidal regions. Across the bootstrap experiments, the ellipsoidal method required an average fitting time of approximately 115 ms per category, compared with approximately 10 ms for the Mahalanobis baseline, corresponding to an approximately 11-fold increase in fitting time. Interestingly, this increase is substantially smaller than the number of regions being fitted: the proposed method typically constructs around 30-40 ellipsoids per category, yet fitting time increases by only around one order of magnitude rather than proportionally with the number of ellipsoids.
A similar but smaller difference is observed during evaluation, where the ellipsoidal method requires approximately 10.3 ms compared with 1.6 ms for Mahalanobis, corresponding to an approximately 6.5-fold increase. Despite these relative increases, the absolute runtime of both fitting and evaluation remains low.

Although the ellipsoidal approach does not outperform the Mahalanobis baseline overall, it provides additional information about how each test sample relates to the learned normal representation. Rather than producing a score relative to a single global distribution, each sample is evaluated against multiple local normal regions. This makes it possible to identify the region to which the sample is most closely related and examine its distance from the surrounding ellipsoidal boundaries. As a result, anomalous samples can be interpreted in terms of their relationship to specific local structures within the normal embedding space, information that is not available from a single global scoring model.

5. Discussion

As discussed in the Results, the proposed ellipsoidal approach does not outperform the Mahalanobis baseline overall. The additional flexibility introduced by modelling multiple local regions of the normal embedding space therefore does not, by itself, translate into improved anomaly-detection performance. However, this does not necessarily indicate that local modelling is ineffective. Performance may instead depend on how regions are constructed, how covariance is estimated within them, and how local regions are combined to represent the overall normal distribution.

The current method follows a relatively greedy local construction strategy, where regions are identified and expanded independently to cover portions of the embedding space. A single global Mahalanobis model represents the opposite extreme, describing the entire normal distribution using one mean and covariance estimate. The stronger baseline performance suggests that the global representation may provide a more statistically stable starting point than constructing the normal space from many independently grown local regions.

The use of a fixed neighbourhood proportion across categories may also contribute to this sensitivity. Although 5% provided a practical compromise during validation, differences in embedding density and structure mean that the same proportion may produce overly local regions for some categories and overly broad regions for others. An adaptive neighbourhood size based on local embedding density may therefore provide a more suitable starting point for region construction.

This motivates an alternative interpretation of the region-growing strategy used in Cloud of Spheres[1]. In its supervised setting, regions can be expanded until they encounter negative samples, providing a natural boundary between classes. In the unsupervised setting considered here, only normal training samples are available, meaning that locally grown regions must instead infer suitable boundaries from the structure of the normal embeddings. Rather than beginning with small K-Nearest

Neighbour-based regions and progressively expanding them, the process could therefore be inverted: a large global normal region could first be estimated and then contracted, divided, or refined where the embedding structure indicates excessive variation. Local modelling would then act as a refinement of the global representation rather than a replacement for it.

The dimensionality of the embedding space presents an additional challenge. DINOv2 ViT-S/14[2] produces 384-dimensional embeddings, while the ellipsoids formed by the proposed algorithm frequently have much lower effective rank. Dividing the training distribution into many small regions reduces the number of samples available for estimating each covariance matrix and can therefore produce rank-deficient local models. This effect varies between categories because some contain fewer normal training samples than the embedding dimensionality, while others contain more. Consequently, low effective rank may reflect insufficient local sample support in some cases and genuine lower-dimensional embedding structure in others. Dimensionality reduction, smaller embedding models, or regularised covariance estimation could therefore be investigated to determine whether a lower-dimensional representation improves stability.

The covariance-sharing heuristic described in Section 3.4 further highlights this issue. Ellipsoids containing insufficient samples borrow covariance information from better-supported regions. While this avoids unstable covariance estimates, repeated borrowing may cause several local regions to inherit similar covariance structures. In the extreme, the model can begin to resemble a fragmented global Mahalanobis representation, but without the advantage of estimating that covariance from the full set of normal samples.

A possible future approach would therefore combine both global and local modelling. A large, well-supported ellipsoid could first capture the dominant normal structure, while samples excluded from this region could be grouped using methods such as K-means[12] and represented by smaller local ellipsoids. These secondary regions could estimate their own covariance when sufficiently supported or model only variation not captured by the primary region. Such an approach would retain the ability to describe multiple modes of normal behaviour while introducing local complexity only where the data indicates that it is required.

6. Conclusion

This work investigated whether normal variation in pretrained DINOv2 embedding spaces could be represented using multiple locally fitted ellipsoidal regions rather than a single global distribution. Although the proposed method does not substantially degrade anomaly-detection performance, it ultimately performs worse than the simpler Mahalanobis baseline, achieving a mean AUROC of 0.934 compared with 0.958. Bootstrap evaluation also indicates that the ellipsoidal approach is more sensitive to variation in the available training samples, suggesting that the additional local flexibility comes at the cost of reduced statistical stability.

For applications where accuracy, simplicity, and computational efficiency are the primary requirements, such as automated manufacturing inspection, the global Mahalanobis model would therefore currently be the more practical choice. However, the proposed representation provides information unavailable from a single global score by relating each test sample to specific locally modelled regions of normality. This local context can also be used to interpret why a sample is considered anomalous, by showing how and where it departs from nearby normal structure rather than only reporting its deviation from a single globally fitted distribution. The experiments also reveal several limitations of constructing many small covariance-based regions in a high-dimensional embedding space, including rank-deficient local estimates and reliance on covariance sharing for weakly supported regions.

Rather than suggesting that local geometric modelling should be abandoned, these results point towards a different construction strategy. Future work could begin from a statistically well-supported global representation and introduce local ellipsoidal refinements only where the embedding structure requires them. This could preserve the stability of the global model while retaining the interpretability and multimodal flexibility that motivated the proposed approach.

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